



Numerical Evaluation of Erosion in Pipeline Elbows Using Alternative Multi-Elbow Configurations in Oil and Gas Application

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Abstract: Solid-particle erosion at pipeline elbows threatens the integrity of oil-and-gas transport systems. This numerical study used the Euler–Lagrange discrete phase model and the Finnie erosion model in ANSYS Fluent to compare a single 90° elbow, two 45° elbows, and three 30° elbows for water–sand flow at inlet velocities of 10–40 m/s and particle diameters of 0.0002–0.0005 m. Two output measures are reported: contour plots show the local cellwise maximum wall erosion rate, whereas line graphs show the area-weighted mean wall erosion rate. At the reference condition of 40 m/s and a particle diameter of 0.0005 m, the area-weighted mean rates were 5.18×10^{-5} , 2.89×10^{-5} , and $3.59 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ for the single 90° elbow, two 45° elbows, and three 30° elbows, respectively. Relative to the single elbow under the same simulation conditions, the mean erosion rate decreased by 44.2% with two 45° elbows and by 30.7% with three 30° elbows. The corresponding local contour maxima were 3.17×10^{-3} , 2.82×10^{-3} , and $2.59 \times 10^{-3} \text{ kg m}^{-2} \text{ s}^{-1}$. These results show that distributing the change in flow direction across multiple elbows reduces severe particle–wall impacts, with two 45° elbows providing the lowest area-weighted mean erosion rate.

Keywords: Pipeline elbow erosion; Oil and gas pipelines; ANSYS Fluent; Solid particle erosion

1 Introduction

Because of erosion brought on by solid particles in turbulent flows, especially in elbows and fittings. The combined impacts of flow and particle factors that significantly affect elbow erosion, emphasizing efficient control methods to extend equipment life [1]. The inner wall of downstream components will be impacted by these particles and weakened, with a rapid change in flow direction causing the most damage to the elbow pipe, erosion can cause the wall thickness to decrease and the strength to deteriorate. The erosion properties of the elbow pipe in a gas-steam ejection system must therefore be investigated [2]. Based on the theory of failure analysis to decide whether the component is suitable to be repaired or not used anymore. The failure is due to corrosion erosion which is an extreme failure and has broken this elbow in just one year. These damage cases should be very detrimental. There is an operating delay, repairs require new spare parts and energy to think about and find the cause of failure to avoid similar events then the causes damage occurs [3].

Pipeline elbow erosion is a problem that must be solved with a scientific and engineering solution, in all dimensions. On the material side, it means the selection of materials which are not erodible and coatings. The fluid side of the effort includes using the elbow geometries and the incorporation of flow control devices. Similarly, optimistically viewing elbows, especially critical service elbows, is important for any long-term plan to reduce and eliminate potentially dangerous pipeline elbow abrasion [4]. Much work has been done to measure and describe the mechanisms of erosion in the elbow, with one of the most notable pieces being by Ada [5]. In that research, high fidelity computational fluid dynamics (CFD) analyses were used to explore the erosion–corrosion occurring in 90° elbow piping systems. The study showed that there is a marked gradient of velocity in the flow in a 90° elbow, and this leads to unequal erosion rates at different locations.

A related study conducted by Song et al. [6] identified several factors that contribute to pipe erosion, including fluid velocity, the size of solid particles, and the smoothness of the pipe's interior surface. The authors of this study believe that velocity is the most critical factor. Their optimal pipe design could reduce the maximum feasible erosion rate by 52.4%. Another study by Ma et al. [7] also looked at structural factors but focused on pipe elbow parameters. They not only looked at the diameter ratio but also at the bend angle, which they found to be a significant factor that better parallels the effect of a geometry-shaping corrugated pipe elbow, leading them to advocate for that elbow type to be used instead of a 90-degree smooth elbow. This is supported by Liu et al. [8] suggested a spiral pipeline structure to reduce the erosion by 34% in similar working conditions.

Khan et al. [9] quantified the combined effects of particle size and flow velocity on elbow erosion; for 300 μm particles, the reported erosion rate at 23 m/s was approximately 3.23 times that at 15 m/s. Jing et al. [10] further related erosion to discharge, sand content, particle shape, and the spacing between consecutive elbows using an Euler–Lagrange framework. Rajkumar et al. [11] examined pipe-size effects experimentally and computationally in liquid-dominated multiphase elbow flows. Under the liquid–solid conditions considered in that study, increasing the pipe diameter from 50.8 to 76.2 mm substantially reduced erosion and altered the spatial erosion pattern; therefore, the pipe-size effect should be interpreted together with the operating and multiphase-flow conditions.

Wang et al. [12] carried out a series of simulations using CFD to study the erosion of π -shaped pipes. They compared different pipe orientations—vertical, horizontal, and 45-degree angles to determine which was most resistant to erosion. Surprisingly, they found that the angled pipe eroded the most. Even more surprisingly, flow velocity and direction affected not just the rate of erosion but the erosion pattern as well higher flow velocities and certain flow directions caused the pipes to “wear” more and to “wear” in a certain way.

Khan et al. [13] compares the elbow orientations under liquid-gas-sand annular flow conditions. They show that the horizontal-vertical (H-V) elbows erode faster than horizontal-horizontal (H-H) elbows, as the H-V elbows erosion rate is 21% higher than the H-H elbows. In their study, Taherifard and Elistratov [14] also found elbow orientation–specific erosion, where more erosion occurred at the second elbow in the vertical pipes and more erosion occurred at the first elbow in the horizontal pipes.

Chen et al. [15] developed an Artificial Intelligence (AI)-based prediction model for erosion in gas–solid elbow flow. Their study demonstrates the potential of data-driven tools for rapid erosion assessment; it is cited here as an AI-prediction contribution and not as the source of the experimental validation data used later in this paper.

The Euler–Lagrange approach is well established for predicting erosion by solid particles in pipe fittings [16, 17]. Recent independent work by Huang et al. [18] combined simulation and validation for particle erosion in a 90° elbow, while Lin et al. [19] and Li et al. [20] examined velocity and continuous-elbow effects. Njobuenwu and Fairweather [21] emphasized the role of turbulent dispersion in dilute particle-laden bend flows, and Oka and Yoshida [22] developed a practical predictive equation for solid-particle impact erosion. Veiskarami and Saidi [23] recently evaluated alternative geometries to a standard elbow. Historically, Edwards et al. [24] established an early CFD framework for solid-particle erosion in elbows and plugged tees.

CFD provides detailed information on flow behaviour, particle trajectories, and erosion patterns that is difficult to obtain from experiments alone [16–20]. Such simulations support erosion-mechanism interpretation and geometry screening, particularly when validated against measurements. Recent independent studies have applied these methods to standard and alternative elbow geometries [18, 23], whereas Ref. [24] is retained specifically for its historical contribution to early CFD erosion modelling.

This research takes a new approach to controlling pipeline erosion. The authors focused on the erosive forces resulting from a combination of two or more pipe bends of varying angles, rather than on the many factors that could influence the erosive forces in any one bend, which have been the subject of other studies. They did so by using characterizations of the different turbulent flow conditions and a series of special dual-elbow designs, from which they made some interesting conclusions. Under a specific set of conditions, the study indicated that a double and triple elbow configuration may not affect corrosion as adversely as a single elbow configuration and that the double and triple elbow may be a good candidate for a design that could benefit in non-competitive, deposited and conveyor system applications.

2 Computational Fluid Dynamics Modeling

2.1 Problem Identification and Boundary Conditions

The work considers the problem of abrasion wear on a vertical pipe with 90°, 45°, and 30° elbow angles under a flow of water and sand as shown in Figure 1. The CFD software ANSYS-Fluent was used to study the effect of the various parameters on erosion, such as fluid velocity and particle diameter. The fluid employed in all the investigations was water with a density of 1000 kg/m³ and velocities of flow varying from 10 m/s to 40 m/s. The pipe, fabricated of mild steel with a diameter of 100 mm and a mass density of 7850 Kg/m³ was exposed to sand particles ranging from 0.0002 m to 0.0005 m with a mass density of 2650 Kg/m³. The radius of curvature is 150 mm, giving an radius-to-diameter (r/D) ratio of 1.5. At the inlet of the pipe, solid particles are uniformly injected at

a velocity of 14 m/s. All of the injected particles are spherical. The no-slip wall condition is applied to the domain walls, which are considered to be perfectly smooth.

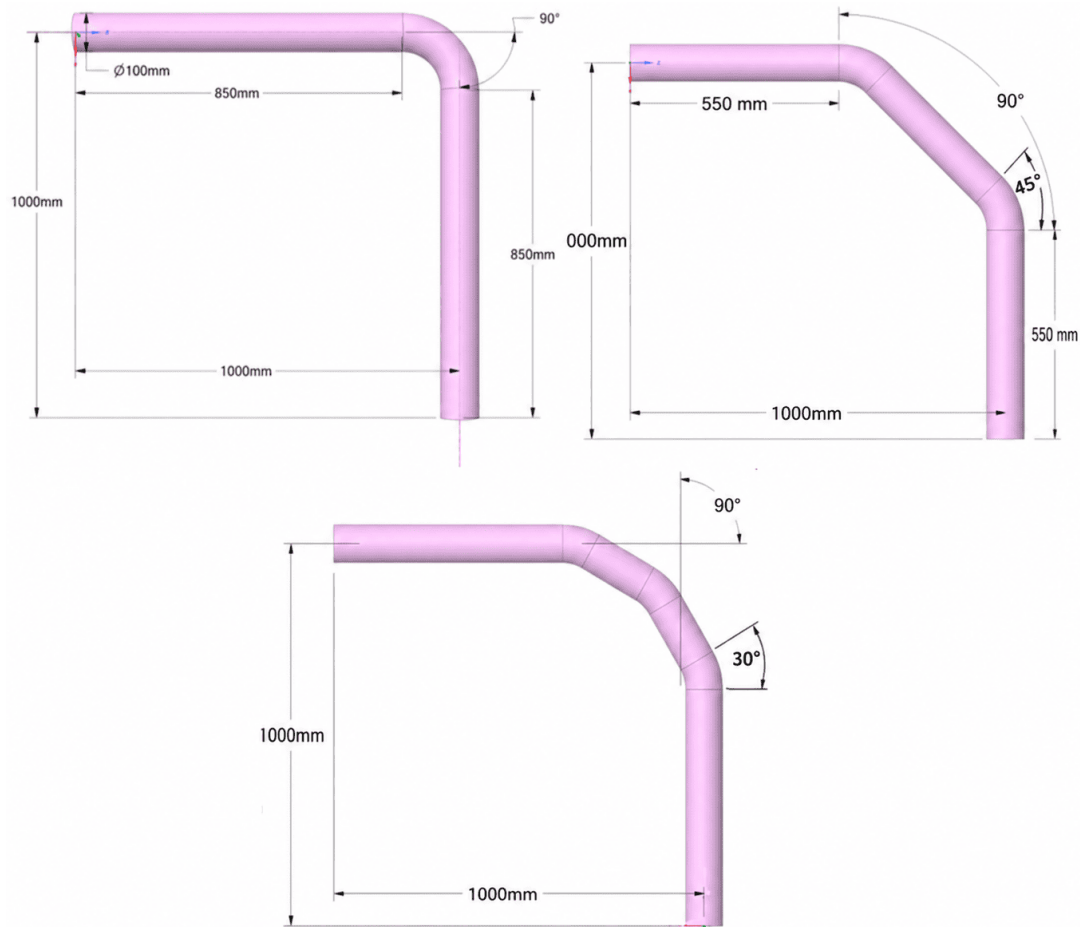


Figure 1. Schematic of 90°, 45°, and 30° vertical pipe elbow

2.2 Mesh Model

The preparation of a computational domain for a CFD simulation in ANSYS 23 is described. The geometry, representing the flow domain, was first created in the Fluent meshing software. To adequately represent the detailed flow behavior throughout the domain, the geometry was discretized into smaller elements, a process known as meshing. Elements of the polyhedral type were chosen, both for their ability to adapt to complex geometry and for the high quality of the resulting mesh, the near-wall region, where there are high-velocity gradients and a boundary layer, requires gradual refinement, as shown in Figure 2.

2.3 Grid Independence Test

A grid-sensitivity test was performed for the single 90° elbow using mesh sizes from 20 to 4 mm (Figure 3 and Table 1). The predicted erosion rate changed non-monotonically: it increased from $5.14879 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ at 20 mm to a maximum of $5.25632 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ at 7.5 mm, and then decreased to $5.15339 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ at 4 mm. Likewise, the successive deviation was non-monotonic and rose to 1.74% at 5 mm before falling to 0.18% at 4.5 mm and 0.04% at 4 mm. The 4 mm mesh contained 619,562 elements. Because the 5 mm result differed from the 4 mm result by only 0.22% while requiring substantially fewer elements, the 5 mm mesh was selected as a practical accuracy–cost compromise; this choice is not presented as evidence of monotonic convergence.

Accordingly, the 5 mm grid containing 365,200 elements was used for the remaining simulations as a practical accuracy–cost compromise.

2.4 Model Validation

The present CFD model was evaluated against the original elbow-erosion measurements reported by Vieira et al. [25] for a 76.2 mm-diameter elbow in gas–solid flow. That experiment used 150 and 300 μm sand particles at gas

velocities of 11–27 m/s. Chen et al. [15] subsequently used data-driven methods for erosion prediction but are not the source of these experimental measurements.

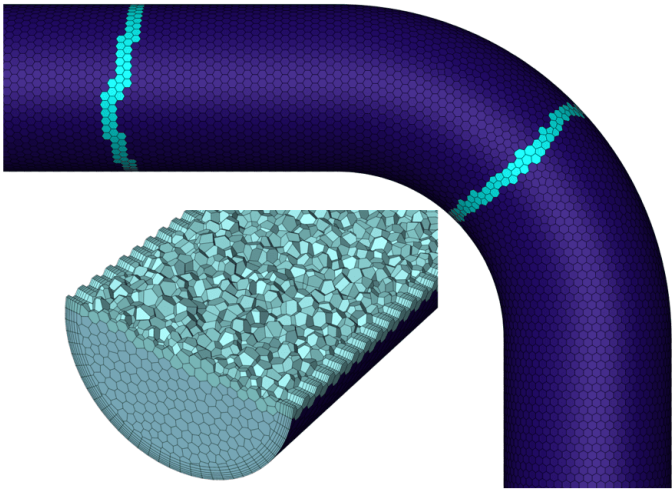


Figure 2. Computational mesh used in the simulation

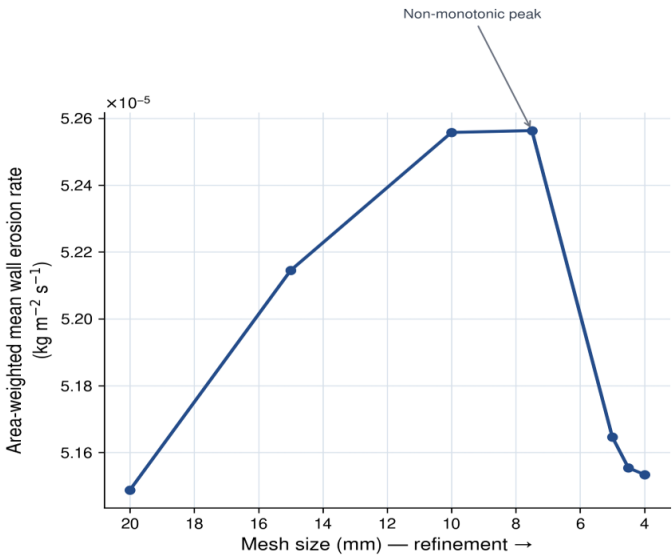


Figure 3. Convergence of erosion rate with grid refinement size

Table 1. Grid independence data for erosion rate prediction

Mesh Size (mm)	Number of Elements	Erosion Rate ($\times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$)	Deviation From Previous (%)
20	34,191	5.14879	-
15	40,635	5.21452	1.28
10	85,224	5.25577	0.79
7.5	158,490	5.25632	0.01
5	365,200	5.16462	1.74
4.5	469,539	5.15540	0.18
4	619,562	5.15339	0.04

Note: (-) = not applicable.

Figure 4 compares the present predictions with the experimental data reported by Vieira et al. [25]. The CFD results reproduce the measured increase in erosion ratio with gas velocity and particle size; the 300 μm particles produce greater erosion than the 150 μm particles. Across the plotted validation points, the difference between the present predictions and the experimental values remains within approximately $\pm 10\%$.

This agreement supports use of the Euler–Lagrange framework with the Finnie model for comparative erosion predictions over the flow conditions considered here.

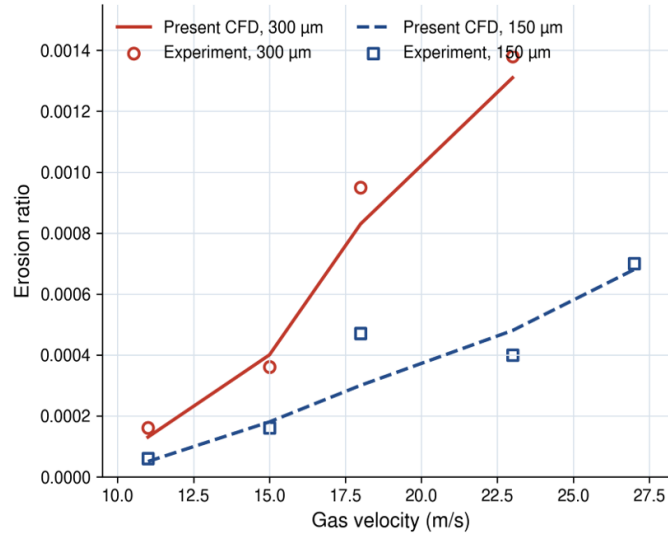


Figure 4. Comparison of the present computational fluid dynamics (CFD) predictions with experimental elbow-erosion data adapted from Vieira et al. [25]

Note: The figure was completely redrawn by the authors from the reported numerical data; no source artwork was reproduced.

2.5 Numerical Procedure

The Navier-Stokes equations govern the motion of the continuous phase (the fluid). Particles (the discrete phase) are treated with Newton's second law. Two-way coupling in motions is implemented in the computations between the continuous and discrete phases. In the assessment of erosion, the Finnie erosion model was applied with the standard k - ϵ turbulence model. The Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) algorithm is employed to link pressure and velocity to enhance convergence behaviour. Second-order upwind differencing is used for convective and divergence terms, while standard discretization methods are used for pressure terms. For all calculations, the overall criteria for convergence are defined as that the residual in the control volume for each equation is less than 10^{-5} or the iterations are equal to 700 in the case of steady simulation.

2.6 Multiphase Model

In the Euler–Lagrange approach for multiphase flow, particularly when dealing with a binary phase flow where one phase is liquid (continuous phase) and the other is solid (discrete phase), the governing equations include the continuity and momentum equations for both phases. Here's a detailed explanation:

Continuity equation for the continuous phase for liquid and particles

The conservation of mass in the continuous phase for liquid is guaranteed by the continuity equation.

$$\frac{\partial \rho_f}{\partial t} + \nabla \cdot (\rho_f u_f) = 0 \quad (1)$$

where, ρ_f : density of the continuous phase (liquid), u_f : velocity of the continuous phase, and t : time.

For the particle phase, the continuity equation:

$$\frac{\partial \rho_p}{\partial t} + \nabla \cdot (\rho_p u_p) = 0 \quad (2)$$

where, ρ_p : density of particle, and u_p : velocity of the particle.

Navier-Stokes equation (momentum conservation) for fluid and particles

The continuous phase (liquid) momentum equation comes from the second law of motion and states that a body of mass M will have an acceleration if a net external force F acts upon it. When we consider an infinitesimal control volume, we can relate the forces acting upon the control volume to the liquid body contained within it. In this way, it can derive the equation for the momentum of the liquid within the control volume.

For the fluid phase, the momentum equation is:

$$\rho_f \left(\frac{\partial u_f}{\partial t} + u_f \cdot \nabla u_f \right) = -\nabla p + \mu_f \nabla^2 u_f + F_{\text{ext}} \quad (3)$$

where, u_f is the velocity of the fluid, p is the pressure in the fluid, μ_f is the dynamic viscosity of the fluid, and F_{ext} represents external forces acting on the fluid (such as gravity).

For the particle phase, the momentum equation is:

$$d u_p / d t = (F_{\text{drag}} + F_{\text{gravity}} + F_{\text{turbulent}} + F_{\text{impact}}) / m_p \quad (4)$$

where, F_{drag} , F_{gravity} , $F_{\text{turbulent}}$, and F_{impact} are the drag, gravitational, turbulence-induced, and impact forces acting on a particle, respectively, and m_p is the particle mass. Dividing their resultant by m_p makes Eq. (4) dimensionally consistent with particle acceleration.

The drag force F_{drag} is typically modeled as:

$$F_{\text{drag}} = \frac{1}{2} C_d \rho_f A_p (u_f - u_p) |u_f - u_p| \quad (5)$$

where, C_d is the drag coefficient, A_p is the cross-sectional area of the particle, u_f and u_p are the velocities of the fluid and particle, respectively.

2.7 Erosion Model

In this study, the Finnie erosion model is used to investigate the erosion wear of the pipeline. The model's governing equation can be expressed as follows:

$$\begin{aligned} \dot{E} &= K V^2 f(\varphi) \\ f(\varphi) &= \frac{1}{3} \cos^2 \varphi, \quad 18.5^\circ \leq \varphi \leq 90^\circ \\ f(\varphi) &= \sin 2\varphi - 3 \sin^2 \varphi, \quad \varphi < 18.5^\circ \end{aligned} \quad (6)$$

where, $\dot{E}$ is the local erosion rate, K is a coefficient associated with the erodent–target material pair and the adopted units, V is the particle impact velocity, φ is the impact angle, and $f(\varphi)$ is the Finnie impact-angle function.

For the ductile mild-steel wall under the present water–sand conditions, the velocity exponent was set to $n = 2$, following the velocity-square dependence of the Finnie model [26].

3 Results and Discussion

Figure 5 shows that the flow dynamics change dramatically when the fluid enters the elbow (as the abrupt change in direction changes dramatically in the flow field), increasing the velocity near the inner bend and causing turbulence throughout the elbow region. At relatively low velocities, the particles maintain a closer alignment with the fluid streamlines, a consequence, at least in part, of the relatively low energies and forces acting on and between the particles. As the flow velocity increases, the energy imparted to the particles also increases; in fact, the relationship between energy and velocity energy is proportional to the square of the velocity, emphasizing that, at higher flow rates, the kinetic energy of the particles has reached an even higher level than at lower flows. Of course, the flow rate (or velocity) and the kinetic energy of the particles are both directly related to the increase in erosion rate at higher potentials, as shown in Figure 6.

At a fixed velocity, larger particles have greater mass and momentum and deviate more strongly from curved streamlines, increasing the likelihood and severity of impacts on the outer elbow wall. Accordingly, the erosion rate also increased with particle diameter (Figure 7). Erosion increased progressively as the inlet velocity rose from 10 to 40 m/s, with 40 m/s producing the maximum value (Figures 6 and 8).

The simulations examined inlet velocities of 10, 20, 30, and 40 m/s and particle diameters of 0.0002, 0.0003, 0.0004, and 0.0005 m. Throughout this section, contour values denote the local cellwise maximum wall erosion rate, whereas Figures 8 and 9 report the area-weighted mean wall erosion rate over the elbow-wall surface. Both quantities are expressed in $\text{kg m}^{-2} \text{s}^{-1}$ and should not be compared as if they were the same statistic.

Elbows in piping systems are well known to be prone to erosion. It is not hard to see why that is the case. The geometry of the elbow itself is a key factor in the increase of these erosion effects. This is because there is a sharp 90° bend, which creates many flow disturbances resulting in turbulence and pressure loss between the inside and outside walls of the pipe. The complex flow field exposes particles to turbulent eddies and flow separation, causing areas of high erosion. The elbow is the most critical part of the piping system as it is where the highest velocity, particle size, and change of flow direction occurs, and thus it has a higher incidence of erosion than elsewhere in the system.

In conclusion, this research identifies the relationship among fluid velocity, particle size and elbow shape on the amount of erosion in a pipe elbow. The combined effect of these three parameters was to augment or reduce the rate

of erosion in various, at times surprising, ways. Designers and operators of systems carrying slurries should pay serious attention to the sort of fluid dynamics calculations performed in this study to ensure that their systems are not too closely mimicked by the unfortunate test conditions of this elbow.

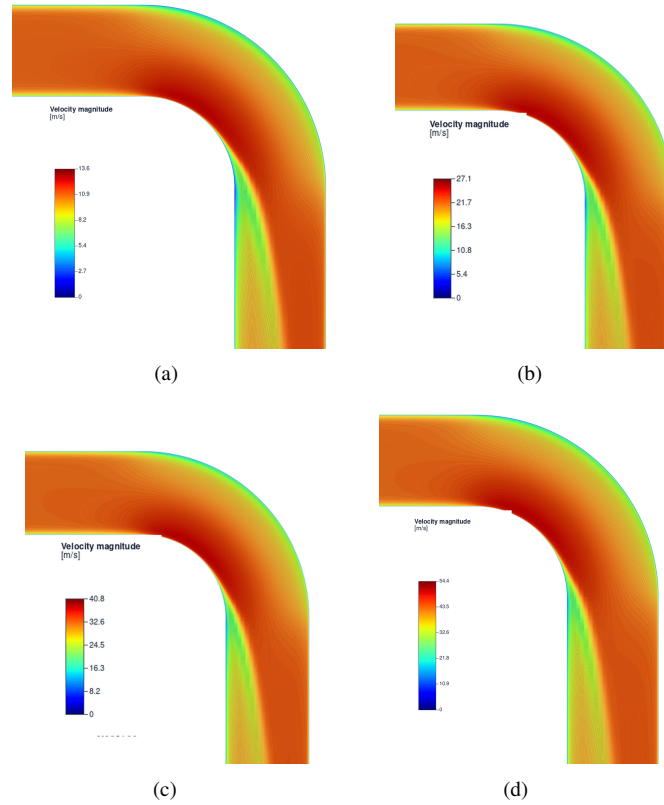


Figure 5. Mid-plane velocity-magnitude contours for the single 90° elbow at inlet velocities of: (a) 10 m/s; (b) 20 m/s; (c) 30 m/s; (d) 40 m/s

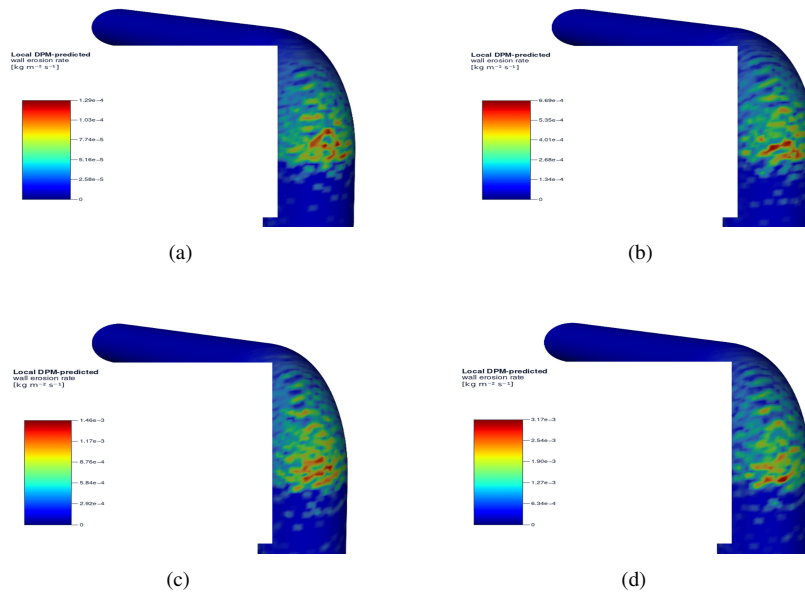


Figure 6. Local discrete phase model (DPM)-predicted wall erosion-rate contours for the single 90° elbow at inlet velocities of: (a) 10 m/s; (b) 20 m/s; (c) 30 m/s; (d) 40 m/s

Note: Each colour bar reports the local cellwise value in $\text{kg m}^{-2} \text{s}^{-1}$.

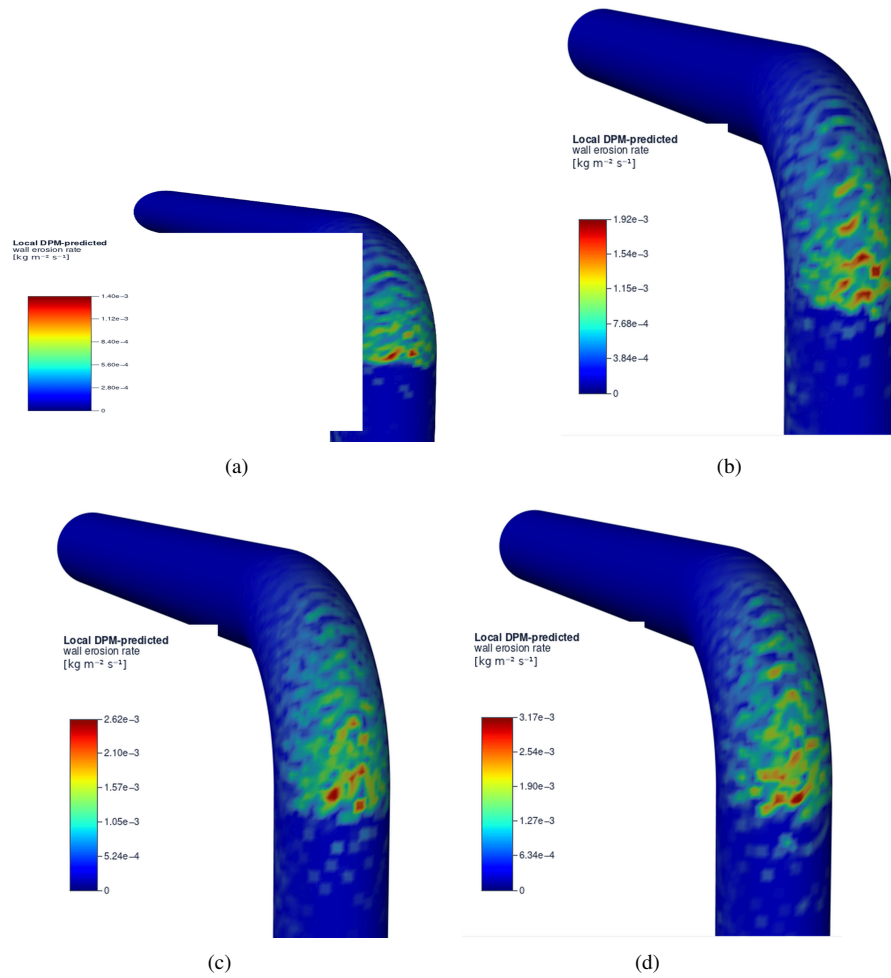


Figure 7. Local discrete phase model (DPM)-predicted wall erosion-rate contours for the single 90° elbow at 40 m/s and particle diameters of: (a) 0.0002 m; (b) 0.0003 m; (c) 0.0004 m; (d) 0.0005 m
 Note: Each colour bar reports the local cellwise value in $\text{kg m}^{-2} \text{s}^{-1}$.

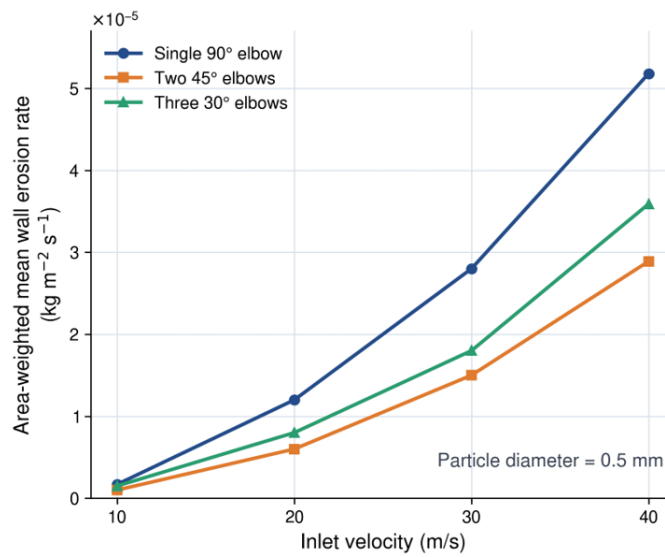


Figure 8. Area-weighted mean discrete phase model (DPM)-predicted wall erosion rate versus inlet velocity for the single 90° elbow, two 45° elbows, and three 30° elbows at a particle diameter of 0.0005 m

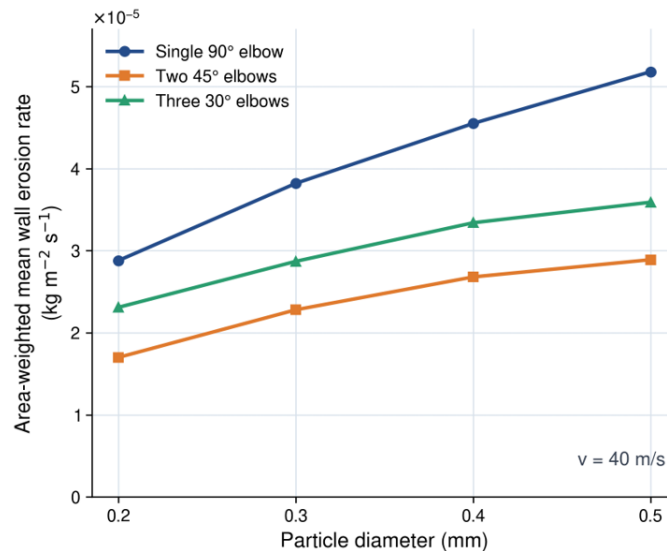


Figure 9. Area-weighted mean discrete phase model (DPM)-predicted wall erosion rate versus particle diameter for the single 90° elbow, two 45° elbows, and three 30° elbows at an inlet velocity of 40 m/s

3.1 Effect of Multiple Elbows

The single 90° elbow was compared with two 45° elbows and three 30° elbows to determine whether distributing the total 90° direction change reduces erosion. The same four particle diameters (0.0002–0.0005 m) and the same inlet velocities (10–40 m/s) were simulated for all three configurations.

The first study involved replacing the single 90° elbow with two 45° elbows. This was an attempt to produce a more gradual distribution of the change of flow direction, so that the fluid and particles were able to negotiate the bends without so much turbulence or sharp impacts with the pipe walls. This smoother flow transition (for particle diameter of 0.0005 m, and same range of velocities used (10, 20, 30, and 40 m/s)) effect was measured and compared with the erosion rate for 90° elbow scenario.

Figure 8 shows that the area-weighted mean erosion rate increases with inlet velocity for all three configurations. At every common velocity, however, two 45° elbows produce a lower mean erosion rate than the single 90° elbow. The gradual direction change promotes smoother particle trajectories, reduces the normal component of particle impact velocity, modifies the impact angle, and limits severe particle–wall collisions; the geometry does not directly reduce particle inertia.

The two 45° elbows also reduce abrupt flow separation and the strength of local turbulent structures relative to a single 90° turn. Consequently, particle–wall impacts are less severe at a given velocity, although erosion still increases as the velocity increases.

After examining the effect of velocity, the effect of various particle diameters on erosion in the modified pipe configuration (with two 45° elbows in Figure 9) was also investigated. Similar to the velocity study, the particle diameters tested were 0.0002, 0.0003, 0.0004, and 0.0005 m. This was done to see if the more gradual flow changes of the 45-degree elbows would have the same effect of reducing erosion rates, as did the higher velocities.

Within each configuration, the area-weighted mean erosion rate increases as particle diameter increases (Figure 9). At the same particle diameter, both multi-elbow configurations nevertheless remain below the single 90° elbow, and the two 45° elbows give the lowest mean rate. The reduction results from smoother trajectories and fewer high-normal-velocity impacts, not from a decrease in particle inertia.

In the single 90° elbow, particles deviate from the fluid streamlines and strike the outer wall with a larger normal velocity component. Distributing the turn across 45° elbows makes the trajectory change more gradual and reduces high-energy impacts. Thus, the multi-elbow designs reduce erosion relative to the single elbow at each matched particle diameter, even though erosion increases with both velocity and particle size within every configuration.

Next, the study investigated the effect of using three 30° elbows instead of a single 90° elbow. The goal was to determine if further increasing the number of elbows would enhance the reduction in erosion. The simulation results showed that while the use of three elbows provided similar benefits to the two-elbow configuration, the overall erosion rate was higher with three elbows than with two, as shown in Figure 9.

This is due to the combined effect of further bends in the flow path. While each individual elbow helps to smooth the flow transition and minimize the impact on the particles, the more elbows the more points the flow needs to change direction. Particles move through three elbows, which creates more opportunities for the wall collisions,

especially on entering and exiting the elbows. However, although the flow is smoother in this case than with the 90-degree elbow, the number of directional changes is greater, and the erosion rate is slightly greater than in the two-elbow case, due to the increased number of particle-wall interactions with this flow.

Multiple elbows have the effect of reducing the severity of the erosion when compared to a single sharp 90° turn, but diminishing returns is seen as more elbows are added. Compared to the flow through three elbows, there is more complexity and particle impact with the introduction of three elbows, but it also counterbalances the advantages of having fewer sharp elbows. Therefore, a two-elbow configuration is found to be more optimal in reducing erosion as opposed to a three-elbow configuration due to smoother flow transition with fewer total number of particle impacts.

Figure 10 compares the three configurations at the reference condition of 40 m/s and a particle diameter of 0.0005 m. The velocity contours and local wall erosion contours reveal different rankings from the area-weighted mean erosion rates and are therefore discussed separately.

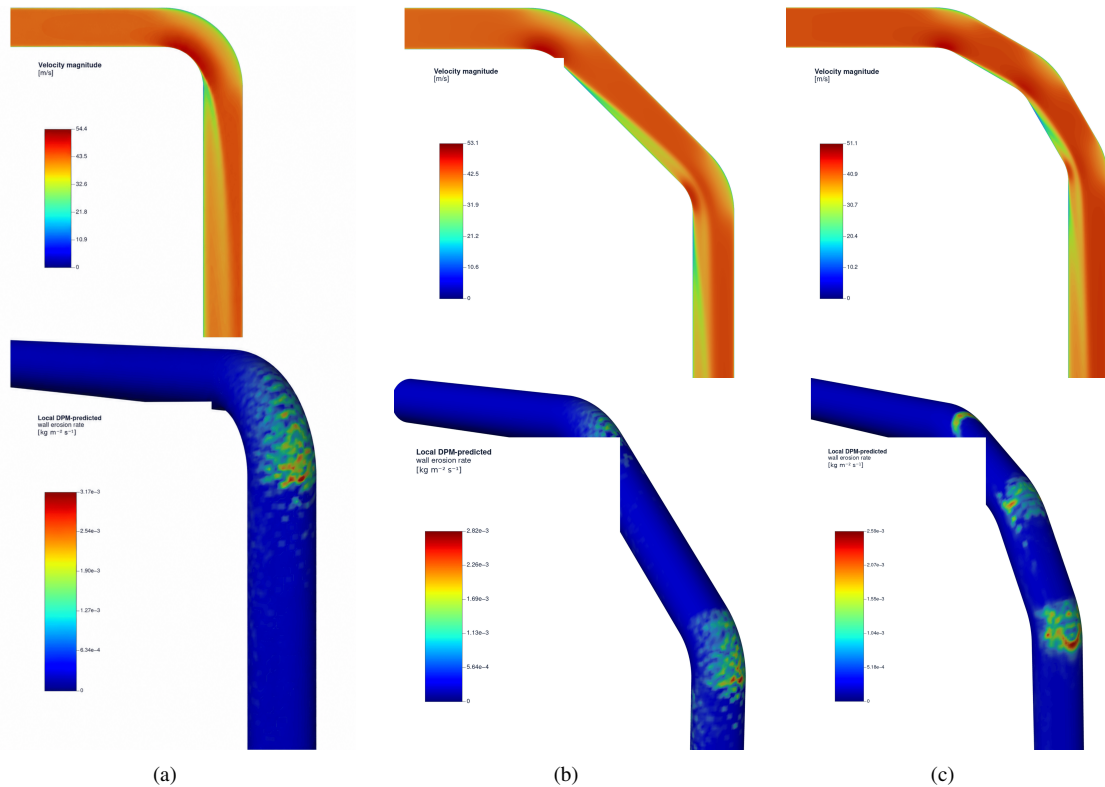


Figure 10. Velocity-magnitude and local discrete phase model (DPM)-predicted wall erosion-rate contours at 40 m/s and a particle diameter of 0.0005 m for: (a) a single 90° elbow; (b) two 45° elbows; (c) three 30° elbows

Note: Colour-bar erosion values are local cellwise rates in $\text{kg m}^{-2} \text{s}^{-1}$.

For the 90° elbow, the velocity contour showed velocities were higher in the elbow region. This is because the fluid accelerates as it turns the corner, due to the sudden change in direction. At the elbow, the sharp bend causes a large rise in velocity, leading to high kinetic energies. This focused flow would raise the erosion potential and increase the effect that the particles can have on the wall, thus increasing the erosion rate, as seen in Figure 10a.

The peak velocity was 54.4 m/s for the single 90° elbow (Figure 10a), 53.1 m/s for two 45° elbows (Figure 10b), and 51.1 m/s for three 30° elbows (Figure 10c). Thus, the three 30° elbows produced the lowest peak velocity, consistent with a more gradual redistribution of the total direction change.

The local cellwise maximum erosion rates in Figure 10 were 3.17×10^{-3} , 2.82×10^{-3} , and $2.59 \times 10^{-3} \text{ kg m}^{-2} \text{s}^{-1}$ for the single 90° elbow, two 45° elbows, and three 30° elbows, respectively. Therefore, the three 30° elbows produced the lowest local maximum. In contrast, the area-weighted mean rates at the same condition were 5.18×10^{-5} , 2.89×10^{-5} , and $3.59 \times 10^{-5} \text{ kg m}^{-2} \text{s}^{-1}$, so two 45° elbows produced the lowest surface-averaged erosion. This distinction resolves the apparent order-of-magnitude and ranking differences between the contour and line plots.

The multi-elbow approach reduced both velocity peaks and erosion relative to the single 90° elbow. Three 30° elbows gave the lowest peak velocity and lowest local maximum erosion, whereas two 45° elbows gave the lowest area-weighted mean erosion rate.

Although two 45° elbows produced the lowest area-weighted mean erosion rate, their local contour maximum exceeded that of the three-elbow configuration. Localized hot spots can arise near the elbow junctions where particle trajectories converge, flow separation persists, or the normal impact-velocity component increases. This spatial concentration explains why a configuration can have a low surface average while retaining isolated high-erosion regions.

4 Conclusion

Changing the elbow configuration reduced erosion in the particle-laden flow simulations. At the reference condition of 40 m/s and a particle diameter of 0.0005 m, the area-weighted mean wall erosion rate was $5.18 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ for the single 90° elbow, $2.89 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ for two 45° elbows, and $3.59 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ for three 30° elbows. These values correspond to reductions of 44.2% and 30.7%, respectively, relative to the single elbow under the same simulation conditions. Three 30° elbows produced the lowest peak velocity and local maximum erosion, while two 45° elbows produced the lowest area-weighted mean erosion. Gradual flow-direction changes can therefore reduce severe particle–wall impacts and may improve pipeline durability.

If erosion is significant, the 2-elbow design is recommended as it offers the best compromise between erosion and a noticeable increase in complexity. However, in the absence of other more extensive flow path modification options, the three-elbow configuration is a viable alternative that can be similarly effective in reducing erosion. Future research could involve studying the influence of varying flow rates, types of fluids and particle properties on the eroded patterns over a wide variety of pipeline configurations to further enhance the design and operation of pipelines resistant to erosion.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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